Evolution of Interdisciplinary Collaboration Patterns in NSF-Funded Research

Zachary Caterer[1,2]

[1]*Department of Chemical and Biological Engineering*
[2]*Interdisciplinary Quantitative Biology Ph.D Program*
University of Colorado Boulder
Boulder, CO, 80303
ztcaterer@colorado.edu

***Abstract*—Interdisciplinary collaboration is a hallmark of modern scientific research, and the National Science Foundation (NSF) has played a critical role in promoting such efforts through its funding mechanisms. In this study we examine how collaboration patterns in NSF-funded research have evolved over the past 25 years by comparing two time periods: 1995-2000 and 2020-2025. Using data from the NSF Awards database, we constructed bipartite networks linking research institutions to grants, which were then projected to create institutional collaboration networks. The results show significant changes in collaborative structures, with modern networks displaying greater connectivity (37% more institutions, 56% more collaborative relationships) while maintaining constant network density. Average path lengths decreased from 3.68 to 3.43, while clustering coefficients increased from 0.340 to 0.417, indicating that institutions now form more tightly interconnected research communities. Analysis of centrality metrics reveals that while established research universities maintain prominence, influence has become more distributed across a wider range of institutions. Funding data shows collaborative grants increased by 46% in number and 155% in funding allocation, reflecting NSF's strategic shift toward supporting interdisciplinary research. These findings suggest the NSF's evolving funding landscape has preserved core collaborative structures while expanding institutional partnerships, potentially facilitating more equitable scientific progress. Future work could explore individual investigator networks, integrate disciplinary metadata, and examine equity dimensions of the collaboration landscape.**

***Index Terms*—Interdisciplinary Collaboration, Network Analysis, Research Funding, Institutional Networks, National Science Foundation (NSF)**

Interdisciplinary Biology Program at the Biofrontiers Institute, University of Colorado Boulder, and the National Science Foundation NRT Integrated Data Science Fellowship (award 2022138)

I. INTRODUCTION

Collaboration is at the core of scientific discovery [1], [2]. Over the past few decades, research has become increasingly interdisciplinary, with scientists joining forces to tackle complex scientific challenges [2], [3]. The National Science Foundation (NSF) has played a central role in facilitating this shift, evolving its funding structures to better support collaborative and cross-disciplinary research [4]. As one of the largest public funders of fundamental research in the United States, the NSF invests over \$8.8 billion annually across a wide range of scientific and engineering fields—from mathematics and computer science to the social sciences and beyond [5].

NSF support comes in many forms, including CAREER awards for early-career investigators and standard research grants for more established projects [6]. It also plays a crucial role in training the next generation of scientists through programs like the NSF Research Experiences for Undergraduates (REU) and the Graduate Research Fellowship Program (GRFP), among many others [7], [8]. In response to the growing complexity of modern science, the NSF has expanded its funding mechanisms to include collaborative research grants and cross-disciplinary solicitations [9]. These structures are designed to foster innovation by bringing together diverse perspectives, skill sets, and institutional resources.

In this study, we investigate the evolution of collaboration in NSF-funded research over a 25-year period by comparing two distinct timeframes:

1995–2000 and 2020–2025. Despite widespread recognition of increasing collaborative trends in science, few studies have systematically quantified how these patterns have transformed at the institutional level. Using network science methods, we address this gap by examining both the structural changes in collaboration networks and shifts in institutional influence within the research landscape funded through the NSF.

Our approach leverages data from the NSF Awards database to construct bipartite networks connecting research institutions to grants. By one-mode projection of these networks onto institutional nodes, we create collaboration networks where edges represent shared grants between institutions, with weights indicating collaboration intensity. This methodology allows us to quantify critical network properties—such as connectivity, clustering, and centrality—providing empirical evidence of how scientific partnerships have evolved in response to changing funding priorities and research influence. Through this analysis, we not only document the changing topology of institutional collaboration but also explore the implications of these structural shifts for scientific innovation, knowledge transfer, and funding policy in an increasingly interconnected research ecosystem.

II. Materials and Methods

A. Data Acquisition and Preprocessing

We retrieved data from the publicly available NSF Awards Database, specifically targeting two five-year periods: 1995–2000 and 2020–2025. For each award, we extracted key metadata, including the participating institutions, funding amounts, grant mechanisms, and research disciplines. To maintain consistency across the dataset, institutions were identified using their unique institution ID. Institutional names were standardized to account for variations in spelling, name changes, and distinctions between campuses (e.g., University of California, Los Angeles versus University of California, Los Angeles Medical School). Consequently, some universities were represented by multiple institutions, reflecting their different campuses or schools.

We made efforts to standardize department names, but due to inconsistencies in the way the information was recorded across grants, some discrepancies remain. For instance, we encountered variations such as "Department of Chemistry" versus "Department of Chemistry and Biochemistry" at the University of Wisconsin-Eau Claire. These inconsistencies were unavoidable given the lack of standardization in the grant metadata.

B. Network Construction

For each five-year period, we constructed a bipartite network in which one set of nodes represented research institutions and the other represented NSF-funded grants. An undirected edge was drawn between an institution and a grant if that institution was listed as a recipient on the award. These bipartite networks were then projected into unipartite (one-mode) networks on the institution layer. In the resulting networks, nodes represent institutions, and an undirected edge between two nodes indicates that the institutions jointly participated in at least one grant. Edge weights denote the number of shared grants between each pair of institutions. Self-loops were not retained to capture intra-institutional collaborations, such as partnerships between departments within the same university are beyond the scope of this analysis. To ensure that subsequent analyses focused on the core collaborative structure, we extracted the largest connected component using NetworkX's `largest_cc()` function [10].

C. Network Metrics

To characterize the structure of each institutional collaboration network, we computed a suite of global and node-level metrics.

At the network level, we calculated the *average path length*, defined as the mean number of steps along the shortest paths for all possible pairs of nodes in the graph. Formally, if $d(u,v)$ is the shortest path length between nodes u and v, and G is connected with n nodes, then the average path length is:

$$L = \frac{1}{n(n-1)} \sum_{u \neq v} d(u,v) \tag{1}$$

We also evaluated the *network density*, which quantifies the proportion of possible edges that are present in the graph. For an undirected network with n nodes and m edges, the density D is given by:

$$D = \frac{2m}{n(n-1)} \quad (2)$$

To assess the tendency of nodes to form tightly knit groups, we calculated the *global clustering coefficient*. This is defined as the ratio of the number of closed triplets (triangles) to the total number of connected triplets of nodes:

$$C = \frac{3 \times \text{number of triangles}}{\text{number of connected triplets of nodes}} \quad (3)$$

At the node level, we analyzed several centrality measures to evaluate institutional influence within the network. *Degree centrality* measures the number of edges incident to a node, capturing the number of direct collaborations an institution has. For node v, the degree centrality is:

$$C_D(v) = \deg(v) \quad (4)$$

Betweenness centrality reflects the extent to which a node lies on the shortest paths between other pairs of nodes, highlighting its role as a potential broker of information or collaboration. It is given by:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (5)$$

where σ_{st} is the total number of shortest paths from node s to node t, and $\sigma_{st}(v)$ is the number of those paths that pass through v.

Finally, we computed *eigenvector centrality*, which assigns relative scores to nodes based on the centrality of their neighbors. It is defined such that a node connected to high-scoring nodes receives a higher score itself. If A is the adjacency matrix of the network and $\vec{x}$ is the vector of centralities, then:

$$A\vec{x} = \lambda\vec{x} \quad (6)$$

where λ is the largest eigenvalue of A.

These metrics jointly provide insight into both the overall structure of inter-institutional collaboration and the relative importance of individual institutions within the network.

D. Null Model Generation

To assess the structural properties of the observed collaboration networks against a randomized baseline, we generated null models using the Watts-Strogatz small-world model. This model preserves certain key characteristics of real-world networks—such as local clustering—while introducing randomness to edge placement, enabling evaluation of whether the observed features deviate significantly from chance.

The Watts-Strogatz model starts with a regular ring lattice in which each node is connected to its k nearest neighbors. Each edge is then rewired with a probability p, introducing randomness while maintaining the network's degree distribution to some extent. The parameters we selected were the number of nodes n, the mean degree k, and the rewiring probability p, set to match the size and density of the observed network.

Formally, the generation process is defined as:

1) Create a ring lattice with n nodes, each connected to $k/2$ neighbors on either side.
2) For each edge (u, v), with probability p, rewire one end of the edge to a randomly chosen node, avoiding self-loops and duplicate edges.

The resulting networks served as null models for comparing key metrics such as average path length, clustering coefficient, and centrality distributions. Repeating this process 1000 times allowed us to construct 95% confidence intervals and assess the statistical significance of structural features observed in the empirical networks.

E. Visualization and Software Tools

All data preprocessing and statistical analyses were conducted using Python and R. Visualizations were generated using Python libraries such as Matplotlib, Plotnine and Seaborn, alongside R's ggplot2 package [11]–[14]. Network construction and visualization were implemented using the NetworkX library in Python [10]. Layouts were optimized for readability using Fruchterman-Reingold force-directed algorithm [10].

III. Results and Discussion

A. Temporal Changes in NSF Funding and Grant Distribution

To understand the shifting landscape of research collaboration, we began by comparing two timeframes from the NSF Awards Database: 1995–1999 (the "historical era") and 2020–2024 (the "modern era").

Figure 4 contextualizes the broader funding environment, showing both the number of grants awarded and total funding distributed across the two eras. Collaborative grants—those involving multiple principal investigators—are highlighted as a subset. As shown in Table I, while the total number of grants increased by 23%, total funding more than doubled, rising by 108%. Collaborative grants exhibited even sharper growth, with a 46% increase in number and a 155% increase in funding.

This disproportionate growth in collaborative funding—outpacing the growth in the number of collaborative grants—suggests a strategic emphasis by the NSF on large-scale, interdisciplinary initiatives. This trend is consistent with the agency's increasing support for convergence research that brings together multiple disciplines to tackle complex societal issues [4].

B. Network Structure Evolution

1) Global Network Properties: To further examine how institutional collaboration has evolved, we constructed networks for each era and compared key metrics (Table II).

The modern network features a 37% increase in participating institutions (nodes) and a 56% increase in collaborative links (edges). When accounting for multiple edges between the same pair of institutions, the increase in collaborative activity is closer to 48%. Despite this expansion, network density remained unchanged at 0.003—suggesting that while the number of collaborations has grown, their distribution remains proportionally consistent. This implies a self-similar scaling in the collaboration network: institutions are joining and forming connections in a manner that preserves the network's overall structure.

Additionally, the average path length decreased from 3.68 to 3.43, indicating that institutions are more tightly interconnected in the modern era. This shift suggests more efficient pathways for knowledge and resource exchange, which could accelerate innovation. Supporting this, the network diameter also decreased from 10 to 9.

One of the most notable changes is the rise in the global clustering coefficient—from 0.340 to 0.417, a 23% increase—indicating stronger local groupings of institutions. Paired with an 8% rise in transitivity, these findings suggest a shift toward denser, multi-institutional collaboration clusters rather than isolated bilateral relationships.

2) Comparison with Null Models: To assess whether the observed network structure could be attributed to random chance, we compared our empirical networks to Watts-Strogatz small-world models calibrated to match each network's size and density. Figure 1 presents this comparison, including 95% confidence intervals from 1,000 simulations.

Both historical and modern networks exhibited significantly shorter average path lengths than their corresponding null models ($p < 0.001$), suggesting that real-world collaborations are more efficient than random associations. While the historical network showed lower clustering and edge counts than expected by chance, the modern network exceeded the null model in edge counts. These results reinforce the conclusion that the evolution of institutional collaboration is not random but reflects increasingly deliberate and structured network formation.

C. Institutional Centrality and Influence

1) Shifts in Institutional Prominence: We next examined which institutions occupy central roles in each era's collaboration network. Table III lists the top five institutions by degree centrality for each era, along with their rankings in betweenness and eigenvector centrality.

Several institutions—like the University of Michigan - Ann Arbor and the University of Colorado Boulder—remain prominent across both periods. However, the scale of their connectivity has shifted substantially. The top-ranked institution in the modern network (Arizona State) shows a 32% increase in degree centrality compared to the historical leader (University of Illinois at Urbana-Champaign), reflecting a general broadening of influence across the network.

TABLE I: Percentage changes in funding allocation and grant numbers between historical (1995-1999) and modern (2020-2024) eras

Category	Funding Increase (%)	Grant Number Increase (%)
Collaborative	155	46
All	108	23

TABLE II: Comparison of network metrics between historical and modern collaboration networks

Network Metric	Historical Era	Modern Era
Number of Nodes	2,353	3,237
Number of Edges	8,668	13,579
Total Edges	12,912	19,129
Network Density	0.003	0.003
Average Path Length	3.68	3.43
Global Clustering Coefficient	0.340	0.417
Network Diameter	10	9
Transitivity	0.167	0.180

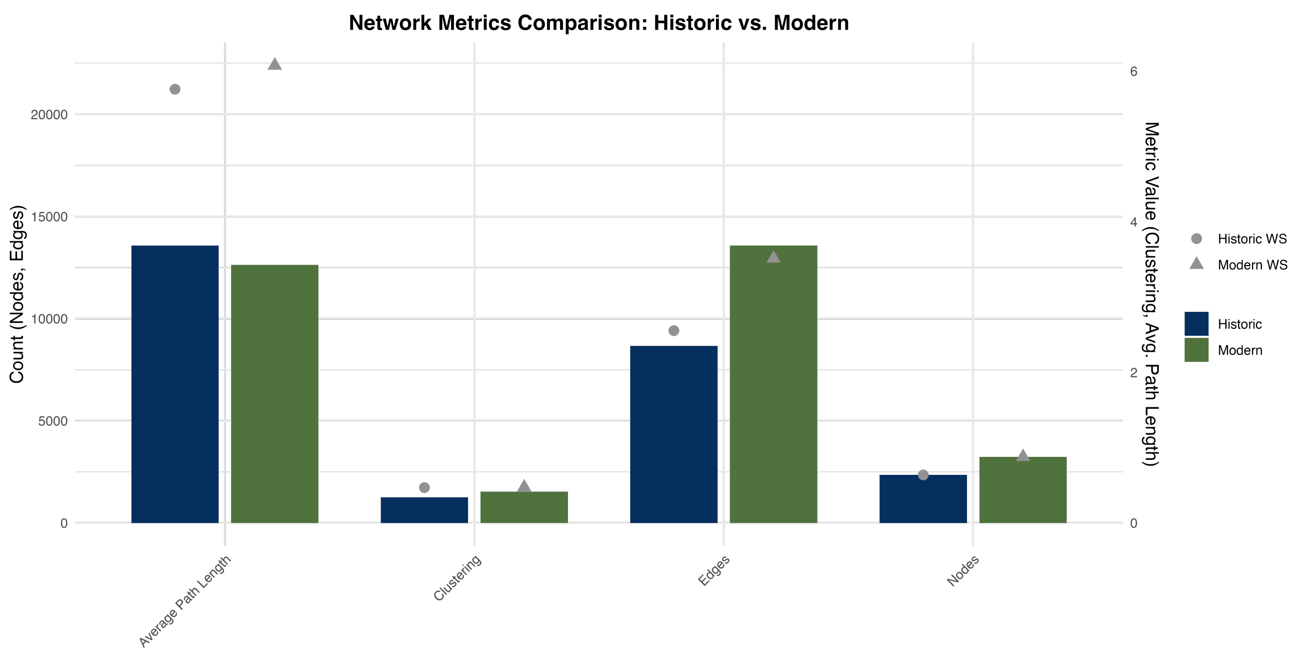


Fig. 1: Comparison of observed network metrics with Watts-Strogatz null model expectations. Error bars represent 95% confidence intervals from 1,000 simulations.

Figure 2 compares the distributions of eigenvector centrality in each era. The modern era displays significantly greater variance ($p < 0.001$, Levene's test [15]), suggesting that influence is more distributed across institutions rather than being concentrated in a few. Interestingly, the average eigenvector centrality did not significantly change ($p > 0.05$, t-test [16]), indicating that while more institutions now hold central roles, the overall level of influence per institution remains consistent.

Together, these findings highlight a transition toward more inclusive and decentralized collaboration networks, where influence is spread across a wider array of institutions.

2) Geographic Diversity: Next, we explored how geographic location influenced institutional collaborations by analyzing the first digit of each institution's ZIP code. We constructed a modularity matrix where each entry represents the frequency of collaborations between institutions grouped by ZIP code digit. This allowed us to examine how spatial proximity shaped collaboration networks over time.

TABLE III: Top five institutions ranked by degree centrality in each era, with corresponding betweenness and eigenvector centrality rankings

Historical Era (1995-1999)				Modern Era (2020-2024)			
Institution	**Deg.**	**Betw.**	**Eigen.**	**Institution**	**Deg.**	**Betw.**	**Eigen.**
University of Illinois at Urbana-Champaign	0.062	5	1	Arizona State University	0.082	1	1
University of Michigan - Ann Arbor	0.061	2	5	Purdue University	0.065	4	2
University of Colorado at Boulder	0.060	4	3	University of Michigan - Ann Arbor	0.064	5	3
University of Arizona	0.060	1	8	University of Colorado at Boulder	0.064	3	12
Pennsylvania State Univ University Park	0.057	6	6	University of Washington	0.063	6	4

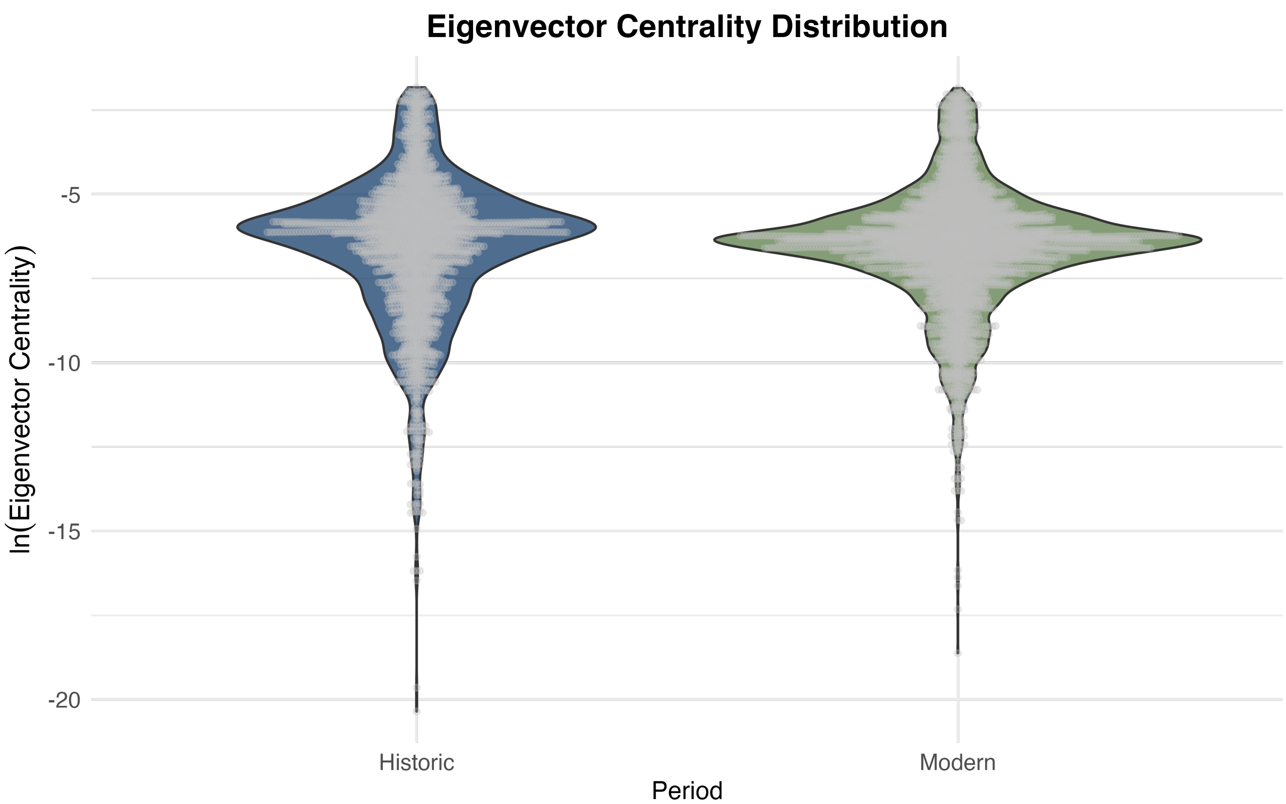


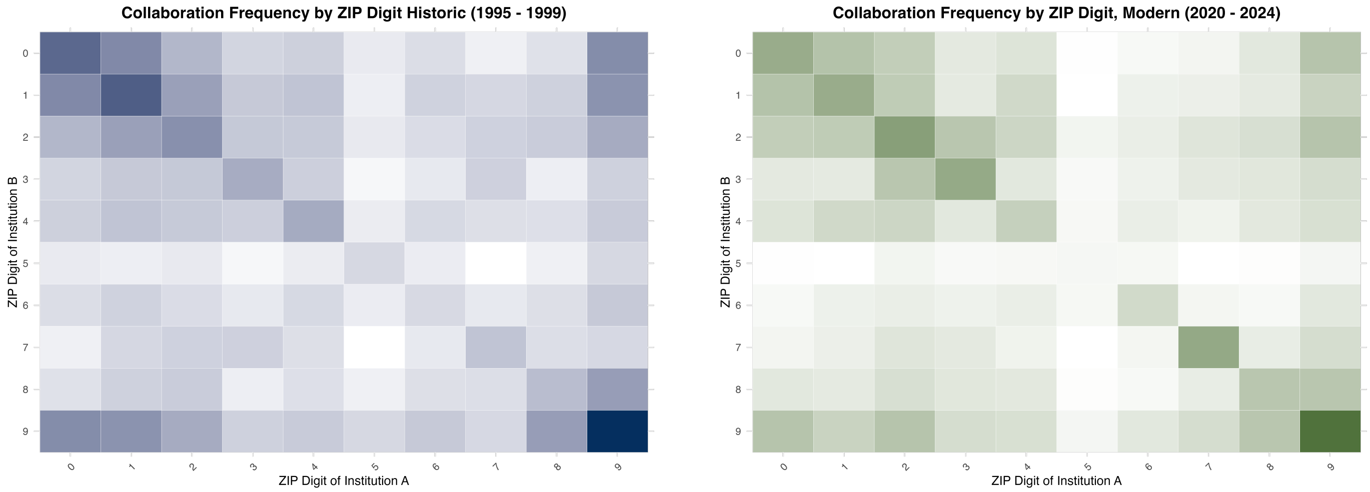
Fig. 2: Distributions of eigenvector centrality for institutions in historical and modern eras. The modern era shows greater variance ($p < 0.001$ [15]). Log scale used for better visualization of the distributions across the eras.

Figure 3 displays these modularity matrices for historic and modern networks.

The historic network shows strong regional clustering, suggesting that collaborations were heavily influenced by geographic proximity. In contrast, the modern network displays a more distributed pattern, with increased collaboration across ZIP code regions. This shift indicates a broader geographical diversity in modern collaborations, likely facilitated by advances in communication and technology which may have facilitated these relationships. While major research institutions continue to play a central role, modern collaborations increasingly include institutions outside traditional geographic strongholds.

D. Implications for Research Funding Policy

Our analysis reveals a fundamental shift in the structure of institutional collaboration networks over the past 25 years. The modern research landscape is marked by denser connectivity, more cohesive communities, and greater interdisciplinary integration. These developments align with the NSF's



(a) Historic ZIP-based collaboration modularity matrix.

(b) Modern ZIP-based collaboration modularity matrix.

Fig. 3: Collaboration frequency by ZIP digit in historic and modern institutional networks.

strategic emphasis on convergence research and reflect evolving approaches to addressing complex scientific challenges.

The rise in funding for collaborative projects relative to single-investigator awards suggests that the NSF has actively incentivized interdisciplinary collaboration. This policy appears to have catalyzed a more interconnected and dynamic research ecosystem, as evidenced by the observed network transformations.

However, increasing stratification in institutional centrality—where established research universities consolidate their dominance—raises concerns about equitable access to collaboration opportunities. While participation by a more diverse set of institutions has improved, the gap between central and peripheral institutions has widened.

These findings suggest that future funding policies should balance excellence with inclusivity. Strategic investment in “bridging institutions” that link otherwise disconnected research communities could enhance knowledge exchange while expanding access to high-impact collaborations.

IV. Conclusions

This analysis of NSF-funded institutional collaborations from 1995–1999 and 2020–2024 highlights major changes in how scientific research is structured and supported in the U.S. Over the past 25 years, there has been a clear shift toward interdisciplinary, multi-institutional projects, with collaborative grants growing much more rapidly than individual ones. This trend reflects national priorities—especially the NSF’s emphasis on convergence research to address complex, interdisciplinary challenges.

Despite the large increase in both the number of institutions and the volume of collaborations, the overall density of the network remained stable. However, other network features suggest the research landscape has become more interconnected and efficient. Shorter average path lengths and higher clustering coefficients point to a system where institutions are more tightly linked and ideas can move more quickly. The fact that these patterns diverge from null models supports the idea that these collaborations are driven by strategic decision-making rather than chance.

Centrality analyses also point to important shifts. While many historically prominent institutions remain central, influence is now more widely distributed across the network. The increased variance in centrality suggests that more institutions are playing important roles, even if the average influence hasn’t changed. This could be a sign of growing opportunities for institutions outside the traditional powerhouses to participate meaningfully in collaborative science.

Taken together, these results suggest that while the core structure of academic collaboration has stayed relatively stable, there has also been a clear expansion in who gets to participate. As science continues to become more collaborative and interdisciplinary, these changes may support broader access to research opportunities and more inclusive progress across fields.

V. Future Directions

While this study provides valuable insights into the evolving structure of NSF-funded institutional collaborations, several promising avenues remain for future investigation. First, expanding the temporal resolution of the analysis to include intermediate time points (e.g., 2000–2009, 2010–2019) could reveal more granular trends in how collaboration structures evolved over time, particularly in response to major policy shifts or technological advancements. Such a longitudinal approach would allow for the identification of inflection points, such as the introduction of specific NSF programs or national initiatives, that may have catalyzed shifts in network topology or institutional centrality.

Second, while this analysis focused on institutional-level collaborations, future work could explore the dynamics at the level of departments, individual investigators, or research teams. Disentangling how personal academic networks grow, fragment, or persist across time could provide a more mechanistic understanding of the social dynamics underpinning institutional collaboration patterns.

Additionally, incorporating topical or disciplinary metadata—such as grant keywords or programmatic themes—would enable the construction of multiplex or multilayer networks. This approach could reveal how different research areas interact, highlighting clusters of interdisciplinary activity and identifying disciplines that serve as bridges between otherwise disconnected fields. Areas such as computer science, data science, mathematics, and statistics are especially likely to function as integrative hubs across diverse domains.

Another critical direction involves examining the equity and inclusivity of the collaboration landscape. For example, mapping institutional collaborations against demographic or geographic indicators may reveal structural disparities in funding access and influence. This would inform efforts to promote a more diverse and equitable research ecosystem.

Finally, coupling this network-based approach with machine learning techniques could enable predictive modeling of collaboration trajectories or funding outcomes. Such tools could help agencies like the NSF proactively identify under-connected institutions with high potential or simulate the effects of policy interventions on collaboration dynamics.

Taken together, these future directions aim to deepen our understanding of the sociostructural mechanisms driving modern scientific collaboration and to inform strategies for optimizing research funding to foster innovation, equity, and collective scientific progress.

VI. Acknowledgments

The authors would like to thank Dr. Daniel Larremore and Dr. Aaron Clauset for their guidance and expertise throughout this project. We are especially grateful to Dr. Clauset for presenting the foundations of network science in a clear and accessible manner. We also acknowledge the BioFrontiers Institute and the Interdisciplinary Quantitative Biology Program for providing this opportunity and for their invaluable support and mentorship.

VII. Code Availability

All code used for this paper and the associated analysis is available at: `github.com/caterer-z-t/CSCI_5352`

In particular, code for parsing the data, generating the network, and performing the network analysis can be found in: `prepare_adj_list.ipynb`

Code used for figure generation is located in: `figures.R`

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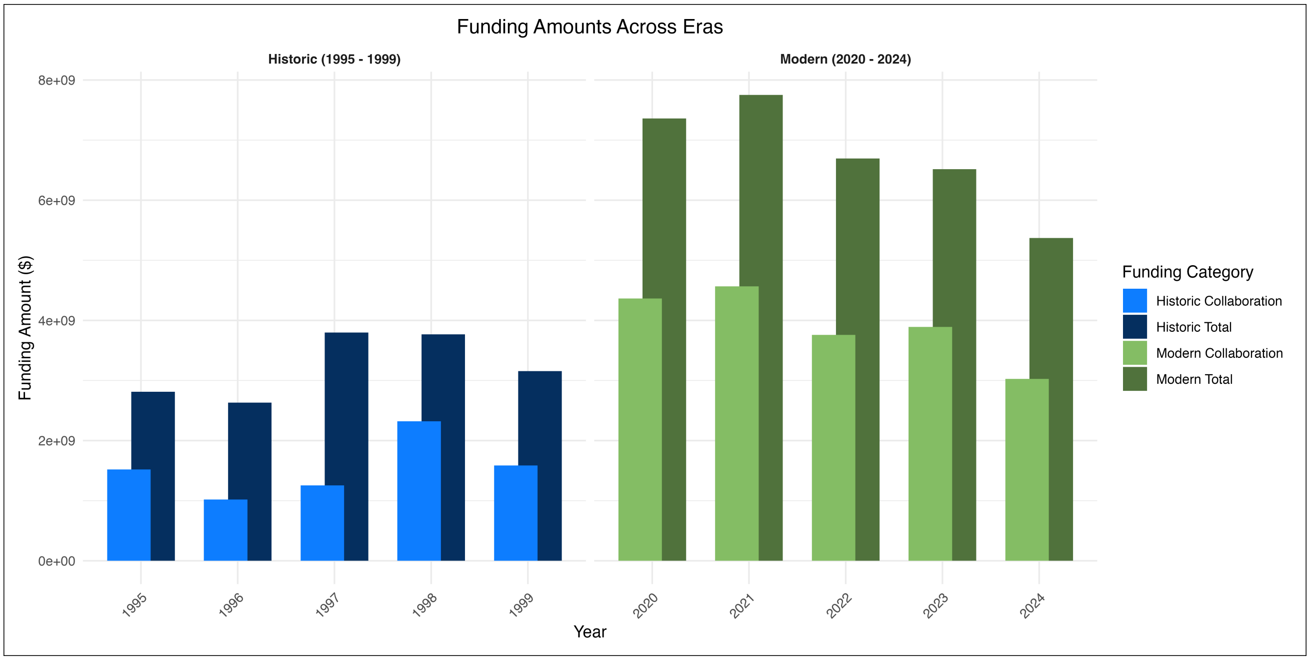
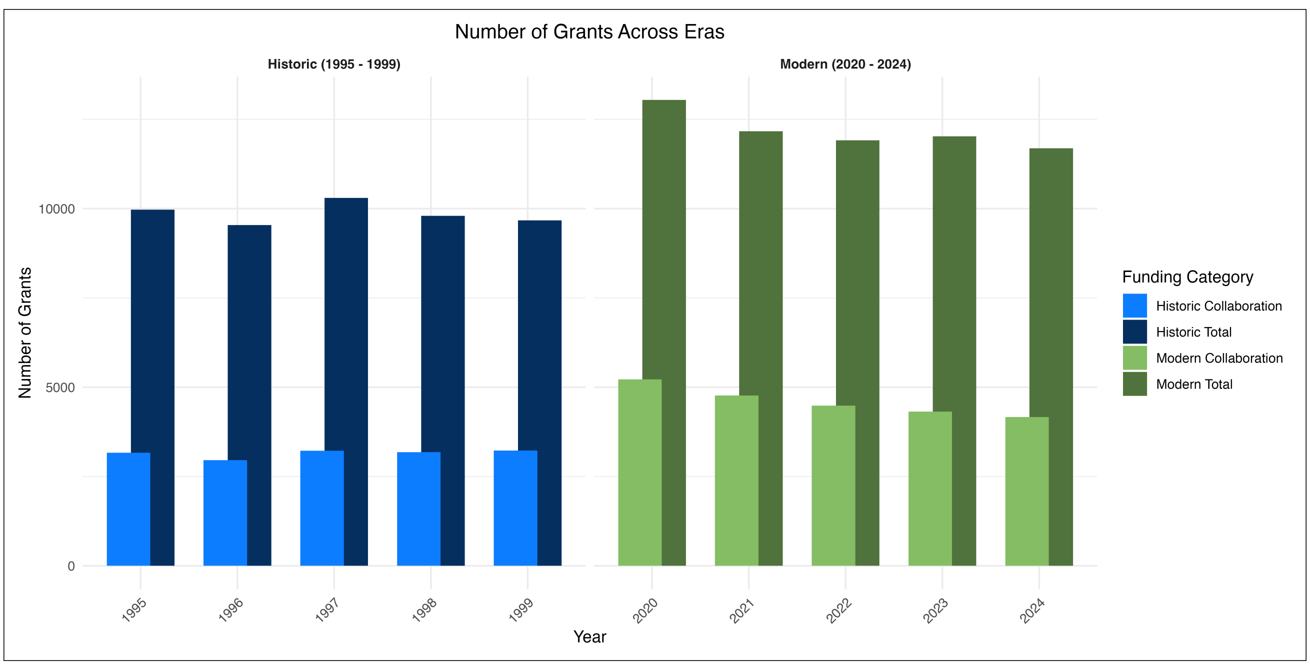
SUPPLEMENTAL MATERIALS



(a) Total funding amounts awarded in collaborative and all NSF grants for both the historic (1995–1999) and modern (2020–2024) periods.



(b) Number of NSF-funded collaborative and total grants awarded during the historic and modern periods.

Fig. 4: Plots comparing total funding amounts (top) and number of grants awarded (bottom) across historic (1995–1999) and modern (2020–2024) periods, with a focus on distinguishing collaborative versus total grants.